Soumodeep Karmakar

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Education

B.E. Electrical Engineering — 2024 – Present
Jadavpur University, Kolkata — *Kolkata, India*

Skills

Languages & Core CS: Python, C/C++, JavaScript, SQL, Object-Oriented Programming, Data Structures & Algorithms
AI & ML: Deep Learning, LLMs, Transformers, Mixture-of-Experts, NLP, Computer Vision, Multimodal Learning, Image Segmentation, Few-Shot Learning, RAG, Agentic AI, Context Engineering
Frameworks & Tools: PyTorch, Hugging Face, OpenCV, scikit-learn, NumPy, FastAPI, Modal, AWS (EC2), Docker, Git

Experience

Research Contributor — Jun 2026 – Present
CMATER Lab, Jadavpur University (under Prof. Ram Sarkar) — *Kolkata, India*

- Built a **cross-dataset, few-shot medical image segmentation** pipeline using **SAM, CLIP, and CoOp** to generalize across unseen medical imaging domains.
- Developed **CLIP-guided mask selection** and pseudo-labeling for domain shift, contributing to **two papers** (one accepted at *CVIP 2026*, one first-author, under review).

Projects

ReAct Agent Framework | *Python, FastAPI, Modal, SWE-bench* |

- Built a modular **ReAct agent framework** with a code-debugging agent (sandboxed bash execution, git patch export) and a chess agent that runs **programmatic lookahead search** inside a sandboxed container.
- Designed a **working-memory context compaction** engine; on SWE-bench `django__django-15368` (resolved) it cut peak prompt size by **86.2%** (35.7K → 4.9K tokens) and total tokens by **86.3%**.
- Added an offline `pytest` suite, retry handling for dropped connections, and strict sandboxing so model-written code never runs on the host.

nanoMoE | 120M params (65.8M active) | *PyTorch, MoE, MLA* |

- Built a **sparse Mixture-of-Experts LLM from scratch** with **DeepSeek-style Multi-head Latent Attention (MLA)** for KV-cache compression, RoPE, and SwiGLU.
- Designed a **Top-2 noisy-gating router** with auxiliary load-balancing loss (four experts per layer); pre-trained on **FineWeb-Edu** using Modal on H100 GPUs.

Kaizen | 72M params | *PyTorch, Transformers* |

- Built a **decoder-only LLM from scratch** in PyTorch: attention, RoPE, RMSNorm, SwiGLU, custom BPE tokenizer.
- Built the training pipeline with **AdamW**, gradient clipping, and cosine LR scheduling.

Flood Semantic Segmentation | *UNet++, ResNet-101, CBAM, PyTorch, SAR+Optical* |

- Multi-class flood segmentation on **SAR + multispectral** imagery with a custom **CBAM adapter** feeding a frozen ResNet-101.
- **Focal + Dice loss** and CutMix for severe imbalance (59 images); **Flood IoU 0.1981**, Top 4 at AISEHack 2026.

Publications

- **Calibrated SAM-CLIP Pseudo-Labeling and Cross-Dataset CutMix for Few-Shot Medical Image Segmentation.** **S. Karmakar**, S. Pani, S. Ghosh, J. Banerjee, R. Pal, R. Sarkar. *Knowledge-Based Systems (Elsevier).* **Submitted (under review)**.
- **Beyond Spatial Prototypes: Joint Spatial–Spectral Prototype Learning for Few-Shot Medical Image Segmentation.** J. Banerjee, R. Pal, S. Ghosh, S. Pani, **S. Karmakar**, R. Sarkar. *CVIP 2026.* **Accepted**.

Achievements

AISEHack 2026 – Top 4 Team, IIIT Hyderabad (IBM Theme: Flood Detection) — Apr 2026
Medha-Manthan 2025 – 1st Prize, IEEE Kolkata Chapter (assistive communication board) — Sep 2025
Hackforge – 3rd Position, Srijan'25, CodeClub Jadavpur University — Mar 2025